

OR Modeling SkillBank Construction

OR Modeling Trajectories

Correct Trajectories



Correct-Error Pairs



LLM-based Skill Distillation



Strategy Distillation

Global Strategy

Problem Type: allocation
Summary: allocate item quantities under bounds
Procedure: define items $\rightarrow$ optimize linear objective $\rightarrow$ enforce bounds



Experience Distillation

Step Experience

Problem Type: allocation
Trigger: quantity may be count-like or fractionally divisible
Guidance: choose variable domains by quantity divisibility

Skill-Augmented OR Formulation

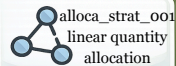
New OR Problem

Step 1: Problem Type Identification

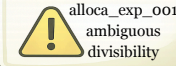
allocation assignment selection
flow routing scheduling
time planning special

Hierarchical SkillBank

Dedup & Merge



Dedup & Merge



Modeling Skill Retrieval

Step 2: Strategy Select

flow_strat_001 flow_strat_002
flow_strat_003 flow_strat_004

Step 3: Experience Select

flow_exp_004 flow_strat_005
flow_exp_005 flow_strat_007

SkillBank Update

Test-Time SkillBank Evolution



LLM Self-Refine



Solver Feedback



Failure Diagnosis

- missing skill
- misleading skill
- code error only
- unresolved

Candidate Skill Pools

Add Skill
for missing-skill case

Repair Skill
for misleading-skill case

Batch-Level Updates

Support Check
adequate coverage

Repair Check
address failures

Stability Check
no performance drop

Reality Label

High

Medium

Low

Risky

Discard
prune risky skill